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MIDDLE VOLTAGE TRANSISTOR AND FABRICATING METHOD OF THE SAME

Abstract

A middle voltage transistor includes a gate formed on a substrate and a gate dielectric layer wider than the gate. First and second spacers are disposed on the gate dielectric layer, with the second spacer having a convex surface. An interface is located between the first and second spacers. A first lightly doped region is formed under the gate, enclosing a second lightly doped region, which in turn encloses a source/drain doped region. A silicide layer covers the source/drain region, with its end located between the convex surface and the interface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. application Ser. No. 17/694,694, filed on Mar. 15, 2022. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present invention relates to a middle voltage transistor which having two lightly doping regions surrounding one source/drain doping region and method of fabricating the same.
2. Description of the Prior Art
[0003] Semiconductor devices are used in a variety of electronic applications such as personal computers, cell phones, digital cameras, and other electronic equipment. The fabricating method of a semiconductor device usually includes sequentially deposit materials of insulators, conductive layers, and semiconductor layers on a semiconductor substrate. Later, materials are patterned to by lithographic processes to form circuit elements on the semiconductor substrate.
[0004] As the integration of semiconductor elements increases, more elements can be integrated into a given area. However, as the feature size shrinks, additional problems arise in each process. For example, when the dielectric layer is removed, the gate dielectric layer is also etched. The etching of the gate dielectric layer leads to current leakage of the transistor completed afterwards.
SUMMARY OF THE INVENTION
[0005] In view of this, the present invention provides a new transistor structure and process so as to prevent current leakage.
[0006] According to a preferred embodiment of the present invention, a middle voltage transistor includes a substrate. A gate is disposed on the substrate. A gate dielectric layer is disposed between the substrate and the gate. A first lightly doping region is embedded in the substrate and extends to be under the gate. A second lightly doping region is embedded within the first lightly doping region, and the first lightly doping region surrounds the second lightly doping region, wherein the second lightly doping region includes a first edge. A source/drain doping region is embedded within the second lightly doping region, and the second lightly doping region surrounds the source/drain doping region, wherein the source/drain doping region includes a second edge. A silicide layer covers and contacts the source/drain doping region, wherein the silicide layer includes an end, and the end is disposed between the first edge and the second edge.
[0007] According to a preferred embodiment of the present invention, a fabricating method of a middle voltage transistor includes providing a substrate. A gate predetermined region is defined on the substrate. Later, a mask layer is formed to cover only part of the gate predetermined region. Subsequently, a first ion implantation process is performed by taking the mask layer as a first mask to implant dopants into the substrate at two sides of the mask layer to form two first lightly doping regions. After removing the mask layer, a gate is formed to overlap an entirety of the gate predetermined region. Next, two second lightly doping regions are respectively formed within one of the two first lightly doping regions. After that, two source/drain doping regions are respectively

formed within one of the two second lightly doping regions. Finally, two silicide layers are formed to respectively cover one of the two source/drain doping regions.
[0008] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** to FIG. **4** depict a fabricating method of a middle voltage transistor according to a preferred embodiment of the present invention, wherein:
[0010] FIG. **1** depicts a substrate having two lightly doping regions thereon;
[0011] FIG. **2** continues from FIG. **1**;
[0012] FIG. **3** continues from FIG. **2**; and
[0013] FIG. **4** continues from FIG. **3**.
[0014] FIG. **5** depicts an enlarged top view of positions of a gate, a first spacer, a second spacer, an interface, a convex outer surface and an end of a silicide layer.
DETAILED DESCRIPTION
[0015] FIG. **1** to FIG. **4** depict a fabricating method of a middle voltage transistor according to a preferred embodiment of the present invention. As shown in FIG. **1**, a substrate **10** is provided. A gate predetermined region A is defined on the substrate **10**. Later, a doping well **11** is formed within the substrate **10**. Subsequently, a mask layer **12** is formed to cover only part of the gate predetermined region A. The mask layer **12** may be photo resist, silicon nitride silicon oxynitride or other materials. In details, the gate predetermined region A has a symmetrical axis S. A symmetrical axis of the mask layer **12** overlaps the symmetrical axis S of the gate predetermined region A. That is, the mask layer **12** is in the middle of the gate predetermined region A. A width W of the mask layer **12** is smaller than the gate predetermined region A. Next, a first ion implantation process **11** is performed by taking the mask layer **12** as a mask to implant dopants into the substrate **10** at two sides of the mask layer **12** to form two first lightly doping regions LDD**1**.
[0016] As shown in FIG. **2**, after the mask layer **12** is removed. A gate dielectric layer **14** is formed to blanketly cover the substrate **10**. Next, a gate **16** is formed on the gate dielectric layer **14**, and the gate **16** covers the entirety of the gate predetermined region A. Now, part of each of the first lightly doping regions LDD**1** is directly under the gate **16**. In other words, part of each of the first lightly doping regions LDD**1** overlaps the gate **16**. It is noteworthy that because the positions of the first light doped regions LDD**1** are defined by the mask layer **12**, the overlapping region between the gate **16** and the first lightly doping regions LDD**1** can be altered by adjusting the width W of the mask layer **12**. Next, two first spacers **18** are formed at two side of the gate **16**. Later, a second ion implantation process **12** is performed by taking the first spacers **18** and the gate **16** as a mask to implant dopants to penetrate the gate dielectric layer **14** to form the second lightly doping regions LDD**2** in the substrate **10**. The second lightly doping regions LDD**2** are disposed at two side of the gate **16**. One of the second lightly doping regions LDD**2** is within one of the first lightly doping regions LDD**1**. The other one of the second lightly doping regions LDD**2** is within the other one of the first lightly doping regions LDD**1**. The first lightly doping region LDD**1** surrounds the second lightly doping region LDD**2** disposed therein. Each of the first lightly doping regions LDD**1** has a range greater than each of the second lightly doping regions LDD**2**.
[0017] As shown in FIG. **3**, two second spacers **20** are respectively formed on one of the first spacers **18**. Next, a third ion implantation process **13** is performed by taking the gate **16**, the first spacers **18** and the second spacers **20** as a mask to implant dopants to penetrate the gate dielectric layer **14** to form two source/drain doping regions S/D in the substrate **10**. The source/drain doping

regions S/D are at two sides of the gate **16**. The source/drain doping regions S/D are respectively embedded within one of the second lightly doping regions LDD**2**. Moreover, the second lightly doping region LDD**2** surrounds the source/drain doping region S/D embedded therein.

[0018] As shown in FIG. **4**, the gate dielectric layer **14** directly on the source/drain doping regions S/D is removed to expose the source/drain doping regions S/D. That is, the part of the gate dielectric layer **14** which is not covered by the gate **16**, the first spacers **18** and the second spacers **20** is removed. When the gate dielectric layer **14** directly on the source/drain doping regions S/D is removed, part of the gate dielectric layer **14** under the second spacers **20** is removed uncontrolledly. Therefore, a concave surface **22** is formed at an end of the gate dielectric layer **14** under the second spacers **20**. This concave surface **22** can even extend to be under the first spacers **18** sometimes. Furthermore, the concave surface **22** recessed in a direction toward the gate **16**. Later, a silicide process is performed to form two silicide layers **24** respectively cover and contact one of the source/drain doping regions S/D. The silicide layers **24** are not only formed directly on the source/drain doping regions S/D, but also fills into the concave surface **22**. That is, the silicide layers **24** will exceed the region directly on the source/drain doping regions S/D. Now, the middle voltage transistor **100** of the present invention is completed. The fabricating process of the middle voltage transistor **100** is suitable for manufacturing N-type transistors and P-type transistors.

[0019] FIG. **4** depicts a middle voltage transistor formed by the fabricating method provided in the present invention. The middle voltage transistor **100** of the present invention can be an N-type transistor or a P-type transistor. The middle voltage transistor **100** includes a substrate **10**. A first direction Y is perpendicular to a top surface of the substrate **10** and a second direction X is parallel to the top surface of the substrate **10**. A gate **16** is disposed on the substrate **10**. A gate dielectric layer **14** is disposed between the substrate **10** and the gate **16**. An end of the gate dielectric layer **14** includes a concave surface **22** recessed in a direction toward the gate **16**. A first lightly doping region LDD**1** is embedded in the substrate **10** and extends to be under the gate **16**. A second lightly doping region LDD**2** is embedded within the first lightly doping region LDD**1**, and the first lightly doping region LDD**1** surrounds the second lightly doping region LDD**2**. The second lightly doping region LDD**2** includes a first edge E**1**. A source/drain doping region S/D is embedded within the second lightly doping region LDD**2**, and the second lightly doping region LDD**2** surrounds the source/drain doping region S/D, wherein the source/drain doping region S/D includes a second edge E**2**. The first edge E**1** and the second edge E**2** are both parallel to the first direction Y.

[0020] A silicide layer **24** covers and contacts the source/drain doping region S/D, wherein the silicide layer **24** includes an end **24***a* and the end **24***a* is disposed between the first edge E**1** and the second edge E**2**. The end **24***a* of the silicide layer **24** contacts the gate dielectric layer **14**. Moreover, a first spacer **18** contacts the gate **16** and is disposed at one side of the gate **16**. The other first spacer **16** also contacts the gate **16** and at the other side of the gate **16**. A second spacer **20** contacts one of the first spacers **16**. The other second spacer **20** contacts the other first spacer **16** which is at the other side of the gate **16**. Each of the second spacers **20** includes a convex outer surface **20***a* that curves away from the gate **16**. Furthermore, an interface **19** is located at a contact region between the first spacer **18** and the second spacer **20** disposed at the same side of the gate **16**.

[0021] The first spacers **18** and the second spacers **20** are all entirely disposed at a top surface of the gate dielectric layer **14**, and a width of the gate dielectric layer **14** is greater than a width of the gate **16**. According to a preferred embodiment of the present invention, an entirety of the silicide layer **24** is not vertically under the first spacer **18**. It is noteworthy that the end **24***a* of the silicide layer **24** is disposed between the convex outer surface **20***a* and the interface **19**. FIG. **5** depicts an enlarged top view of positions of a gate, a first spacer, second spacers, an interface, a convex outer surface and an end of a silicide layer. As shown in FIG. **5**, more specifically, as seen from a top view, the end **24***a* of the silicide layer **24** is disposed between the convex outer surface **20***a* and the interface **19**.

[0022] The width of the gate dielectric layer **14** and the width of the gate **16** are both parallel to the second direction X. The second direction X is not only parallel to the top surface of the substrate **10**, but also extends from one source/drain doping region S/D to the other source/drain doping region S/D. The width of the gate dielectric layer **14** equals to the summation of the width of the gate **16**, the width of the two first spacers **18** and the width of the two second spacers **20**.
[0023] Furthermore, the gate **16** includes a third edge E**3** parallel to the first direction Y, the first lightly doping region LDD**1** includes a fourth edge E**4** parallel to the first direction Y. According to a preferred embodiment of the present invention, along the second direction X, a distance between the third edge E**3** and the fourth edge E**4** is smaller than half of the width of the gate **16**. Moreover, the operational voltage of the middle voltage transistor **100** can be operated at a normal operational voltage or an under drive voltage. For example, the operational voltage of the middle voltage transistor **100** can be between 1.8 and 9 volts. The thickness of the gate dielectric layer **14** is preferably between 100 and 250 angstroms.
[0024] The first lightly doping region LDD**1** has a first dopant concentration, the second lightly doping region LDD**2** has a second dopant concentration, the source/drain doping region S/D has a third dopant concentration. The third dopant concentration is greater than the second dopant concentration, and the second dopant concentration is greater than the first dopant concentration. For example, the first dopant concentration is between 1E17 and 5E18 cm.sup.−3, the second dopant concentration is between 5E18 and 9E18 cm.sup.−3, and the third dopant concentration is between 5E18 and 9E20 cm.sup.−3. The source/drain doping region S/D of the present invention is surrounded by two lightly doping regions (the first lightly doping region LDD**1** and the second lightly doping region LDD**2**). The depth of the first lightly doping region LDD**1** is greater than the depth of the second lightly doping region LDD**2**. The depth of the second lightly doping region LDD**2** is greater than the depth of the source/drain doping region S/D. On the contrary, in a conventional transistor, the depth of the lightly doping region is smaller than the depth of the source/drain doping region S/D.
[0025] According to the fabricating process of the middle voltage transistor **100** mentioned above, the silicide layer **24** is not only formed directly on the source/drain doping regions S/D, but also fills into the concave surface **22** to make the end **24***a* of the silicide layer **24** to extend to be under the second spacer **20**. If there is no second lightly doping region LDD**2**, the difference between the dopant concentration of the source/drain doping region S/D and the dopant concentration of the first lightly doping region LDD**1** will be too large, and electrons will punch through at the concave surface **22**, and then current leakage occurs. Therefore, the second lightly doping region LDD**2** which having a dopant concentration between the dopant concentration of the source/drain doping region S/D and the dopant concentration of the first lightly doping region LDD**1** is specially provided in the present invention. Moreover, the end **24***a* of the silicide layer **24** is in a position which does not exceed the first edge E**1** of the second lightly doping region LDD**2**. In this way, current leakage can be effectively prevented.
[0026] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A middle voltage transistor, comprising: a substrate; a gate disposed on the substrate; a gate dielectric layer disposed between the substrate and the gate, wherein a width of the gate dielectric layer is greater than a width of the gate; a first spacer contacting the gate and disposed at one side of the gate; a second spacer contacting the first spacer, wherein the second spacer comprises a convex outer surface that curves away from the gate, and the first spacer and the second spacer are entirely disposed on a top surface of the gate dielectric layer; an interface located at a contact region between the first spacer and the second spacer; a first lightly doping region embedded in the substrate and extending to be under the gate; a second lightly doping region embedded within the first lightly doping region, and the first lightly doping region surrounding the second lightly doping region; a source/drain doping region embedded within the second lightly doping region, and the second lightly doping region surrounding the source/drain doping region; a silicide layer covering and contacting the source/drain doping region, wherein the silicide layer comprises an end, the end is disposed between the convex outer surface and the interface, and the end contacts the gate dielectric layer.

2. The middle voltage transistor of claim 1, wherein the silicide layer is not vertically under the first spacer.

3. The middle voltage transistor of claim 1, wherein an end of the gate dielectric layer comprises a concave surface recessed in a direction toward the gate.

4. The middle voltage transistor of claim 1, wherein a first direction is perpendicular to a top surface of the substrate, a second direction is parallel to the top surface of the substrate, the gate comprises a third edge parallel to the first direction, the first lightly doping region comprises a fourth edge parallel to the first direction, and wherein along the second direction, a distance between the third edge and the fourth edge is smaller than half of the width of the gate.

5. The middle voltage transistor of claim 1, wherein a thickness of the gate dielectric layer is between 100 and 250 angstroms.

6. The middle voltage transistor of claim 1, wherein an operational voltage of the middle voltage transistor is between 1.8 and 9 volts.

7. The middle voltage transistor of claim 1, wherein the first lightly doping region has a first dopant concentration, the second lightly doping region has a second dopant concentration, the source/drain doping region has a third dopant concentration, the third dopant concentration is greater than the second dopant concentration, and the second dopant concentration is greater than the first dopant concentration.

8. The middle voltage transistor of claim 7, wherein the first dopant concentration is between 1E17 and 5E18 cm.sup.−3, the second dopant concentration is between 5E18 and 9E18 cm.sup.−3, and the third dopant concentration is between 5E18 and 9E20 cm.sup.−3.